

BARCELONA, Venezuela

WMO: 804190

Lat: 10.107N

Lon: 64.689W

Elev: 26

StdP: 14.68

Time Zone: -4.00 (W04)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	70.4	71.7	58.5	73.1	85.1	60.5	78.9	83.6	22.9	89.6	21.6	89.6	1.0	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	17.7	96.4	79.0	95.1	78.7	94.5	78.5	84.3	91.3	82.8	90.2	81.8	89.4	12.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.5	169.7	88.6	80.9	160.6	86.5	80.3	157.3	85.7	48.4	90.7	46.7	90.1	45.6	90.6	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	18.5	16.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	81.8	82.0	83.2	84.0	84.7	83.7	82.7	83.0	83.9	84.0	83.9	82.7
	DBStd	1.81	1.55	1.52	1.70	1.79	1.57	1.77	1.70	1.44	1.68	1.55	1.29	1.55
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12153	984	896	1030	1020	1075	1011	1013	1024	1016	1055	1017	1012
	CDD65	6678	519	476	565	570	610	561	548	559	566	590	567	547
	CDH74	73690	5528	5176	6560	6776	7351	6218	5554	5721	6069	6512	6335	5890
	CDH80	33771	2445	2363	3146	3236	3570	2810	2353	2436	2807	3044	2931	2630
Wind	WSAvg	4.6	4.6	5.3	5.6	6.0	5.1	4.3	3.7	4.0	4.4	4.6	3.7	4.3
Precipitation	PrecAvg	24.0	0.2	0.1	0.2	0.3	1.6	3.6	4.8	4.6	3.0	2.3	2.1	0.9
	PrecMax	36.9	1.6	1.7	1.2	3.3	8.9	8.6	11.6	11.7	8.2	6.1	7.9	2.8
	PrecMin	15.7	0.0	0.0	0.0	0.0	0.0	0.4	2.0	0.6	0.5	0.3	0.2	0.0
	PrecStd	5.2	0.4	0.3	0.3	0.5	1.7	1.7	2.2	2.2	1.5	1.4	1.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	95.2	96.8	96.8	96.9	95.4	94.9	94.9	97.0	96.9	95.4	95.1
		MCWB	77.2	78.8	78.1	80.1	78.4	78.3	78.8	78.7	79.8	79.1	79.6	78.1
	2%	DB	93.0	93.2	94.8	95.0	95.2	93.7	93.1	93.2	95.1	95.0	94.6	93.3
		MCWB	77.2	77.7	77.7	78.2	78.8	78.3	78.3	78.7	78.7	79.0	79.6	77.8
	5%	DB	91.5	91.6	93.2	93.3	93.6	93.0	91.5	91.7	93.4	93.5	93.2	91.7
		MCWB	76.9	76.7	77.5	77.1	78.3	78.2	77.7	78.4	78.8	79.3	79.2	77.2
	10%	DB	89.9	90.0	91.6	91.7	92.9	91.3	89.8	90.0	91.7	91.9	91.6	90.9
		MCWB	76.6	76.0	76.8	76.9	78.1	77.9	77.5	78.0	78.4	78.8	78.8	77.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.5	83.4	84.0	85.2	85.2	84.2	83.0	82.7	83.6	83.9	87.0	82.3
		MCDB	89.5	91.9	91.3	93.6	93.2	90.6	89.5	89.2	90.2	90.8	90.6	89.0
	2%	WB	80.7	80.8	82.0	82.5	83.1	82.1	81.2	80.9	81.5	82.2	82.8	80.7
		MCDB	88.5	88.7	89.9	90.9	91.4	88.3	87.7	87.9	89.9	89.9	89.9	88.0
	5%	WB	79.5	79.3	80.3	80.3	81.4	80.5	80.1	80.0	80.5	81.0	81.1	79.8
		MCDB	87.1	87.1	88.4	88.7	89.5	87.4	86.7	87.1	89.0	89.7	88.8	87.4
	10%	WB	78.5	78.1	79.1	79.1	80.0	79.5	79.2	79.2	79.6	80.0	80.1	78.9
		MCDB	85.8	86.1	87.1	87.3	87.9	86.4	85.7	86.4	87.9	88.7	87.7	86.8
Mean Daily Temperature Range	5% DB	MDBR	18.8	19.2	19.0	17.8	17.7	17.1	16.7	16.9	18.3	18.1	17.9	18.0
		MCDBR	20.0	20.2	20.1	19.0	18.8	18.6	18.0	18.0	19.6	19.5	19.0	19.1
		MCWBR	8.1	8.0	8.0	7.2	7.3	6.5	6.8	6.8	7.2	7.5	7.5	7.2
	5% WB	MCDBR	19.0	19.5	19.0	18.3	18.1	17.2	17.1	17.2	18.7	18.2	18.2	18.4
		MCWBR	9.6	9.9	9.8	8.7	9.2	7.9	7.8	7.6	8.0	8.3	9.1	8.0
Clear-Sky Solar Irradiance	taub	0.379	0.398	0.407	0.459	0.449	0.479	0.451	0.424	0.413	0.393	0.389	0.394	
	taud	2.561	2.495	2.469	2.321	2.376	2.268	2.374	2.468	2.479	2.552	2.569	2.544	
	Ebn at Noon	295	292	290	271	269	258	266	277	284	290	290	288	
	Edh at Noon	32	35	37	42	39	43	39	36	36	33	32	32	
All-Sky Solar Radiation	RadAvg	1704	1870	1951	1907	1830	1774	1815	1895	1927	1813	1659	1602	
	RadStd	63	69	123	102	105	78	77	74	61	55	85	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page